

SIMATS ENGINEERING

ASSESSMENT TOOL 2

Case study

COURSE NAME: Computer Networks **COURSE CODE:** CSA0704

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Case study

QUESTIONS:4 Total Marks:40

1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss**, **180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS**, **DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.

3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

3.A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

- 4. Examine the impact of centralized DNS architecture on application performance.
- 5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
- 6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

- 1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.
- 2. Explain how each factor affects transaction processing in high-frequency systems
- 3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and	Good analysis with logical reasoning	Basic analysis with limited	Weak or no analysis; lacks reasoning

		insights		depth	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q1.Real-Time Video Conferencing

1.Analyze whether the problem is primarily caused by the transport protocol (TCP/UDP) or the application layer.

The problem is primarily related to the **transport protocol and its interaction with the application layer**. Real-time video conferencing normally prefers **UDP** because it provides low-delay transmission without waiting for retransmission of lost packets. The observed **8% packet loss and 180 ms jitter** can cause voice distortion, frozen video, and delayed communication.

If TCP is used for real-time media, packet loss can trigger retransmissions and congestion control, increasing delay. However, the application layer also plays an important role because improper buffering can increase latency when jitter is high.

Therefore, the main issue is the **combination of network packet loss, high jitter, transport protocol behavior, and inefficient application buffering**.

2. Justify your answer based on the communication requirements of real-time multimedia applications.

Real-time multimedia applications require:

- **Low latency** – audio and video must reach users quickly.
- **Low jitter** – packets should arrive at relatively consistent intervals.
- **Low packet loss** – excessive loss produces distorted audio and frozen video.
- **Continuous transmission** – temporary packet loss is generally preferable to waiting for retransmission.
- **Efficient buffering** – buffers should compensate for jitter without introducing excessive delay.

UDP is suitable because it does not wait for lost packets to be retransmitted. For conferencing, receiving a slightly incomplete video frame is often better than

receiving it several seconds late.

The measured **180 ms jitter** and **8% packet loss** are therefore serious for a real-time application.

3. Recommend suitable protocol-level or application-level improvements.

Protocol-level improvements:

1. Prefer **UDP-based real-time media transport** rather than TCP for audio/video.
2. Use congestion-control mechanisms suitable for real-time traffic.
3. Apply **Quality of Service (QoS)** to prioritize voice and video packets.
4. Use packet-loss recovery techniques such as **Forward Error Correction (FEC)** where appropriate.
5. Monitor and reduce network congestion.

Q2.Global DNS Infrastructure

1. Analyze the reasons behind the high DNS resolution time.

The average DNS lookup time is **800 ms**, which is considerably high for a web application. Possible reasons include:

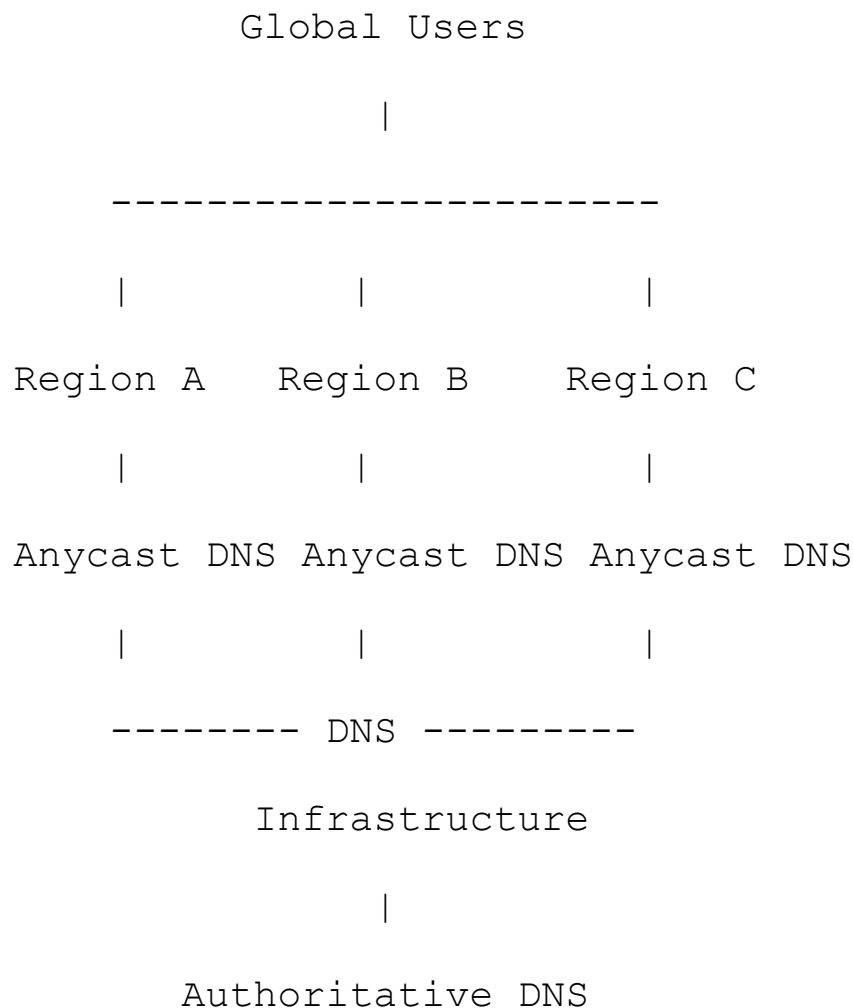
- DNS requests being sent to a distant server.
- Excessive network latency.
- Multiple DNS resolution steps.
- Poor DNS caching.
- Incorrect or very short TTL values.
- Overloaded DNS servers.
- Lack of geographically distributed DNS servers.
- Network congestion between users and DNS infrastructure.

Since DNS resolution occurs before the application homepage can be accessed, an 800 ms lookup directly increases the user's initial page-load delay.

2. Propose a suitable DNS architecture to reduce lookup time below 50 ms.

A **distributed Anycast DNS architecture** should be implemented.

Architecture:



Improvements:

1. **Anycast DNS:** Advertise the same DNS IP address from multiple geographically distributed locations. Users are directed to a nearby DNS server.
2. **DNS caching:** Frequently requested records should be cached to avoid repeated resolution.
3. **DNS pre-fetching:** Frequently required DNS information can be resolved before the user requests the application.
4. **TTL optimization:** Appropriate TTL values allow records to remain cached while still allowing changes to propagate when required.
5. **Redundant DNS servers:** Multiple authoritative servers should be deployed to avoid a single point of failure.

With geographically closer servers, caching, and optimized TTL values, DNS response time can potentially be reduced toward the required **below-50-ms target**, depending on network conditions.

3. Evaluate the advantages and limitations.

Aspect	Advantages	Limitations
Performance	Reduces lookup latency	Requires distributed infrastructure
Scalability	Handles large global traffic	More servers increase management complexity
Availability	Failure of one location can be handled by another	Configuration errors can affect multiple locations
Anycast	Directs users toward nearby DNS infrastructure	Requires suitable routing configuration
Caching	Reduces repeated DNS queries	Stale records can exist until TTL expires
TTL optimization	Balances performance and update speed	Very high TTL can delay DNS changes

Conclusion: A combination of **Anycast DNS, caching, pre-fetching, optimized TTL values, and redundant authoritative servers** provides a scalable and highly available DNS architecture.

Application-level improvements:

- 1. Use an **adaptive jitter buffer**.
- 2. Dynamically reduce video resolution or frame rate when bandwidth decreases.
- 3. Prioritize audio over video during severe congestion.
- 4. Use adaptive bitrate streaming.
- 5. Monitor packet loss, jitter, and available bandwidth continuously.

Conclusion: A combination of UDP-based transmission, QoS, adaptive buffering, and adaptive video quality can significantly improve conferencing quality.

Q3.Centralized vs Distributed DNS

4. Examine the impact of centralized DNS architecture on application performance.

A centralized DNS architecture sends all DNS requests to a **single primary DNS server**. In the given case, DNS queries take approximately **800 ms**.

This creates several problems:

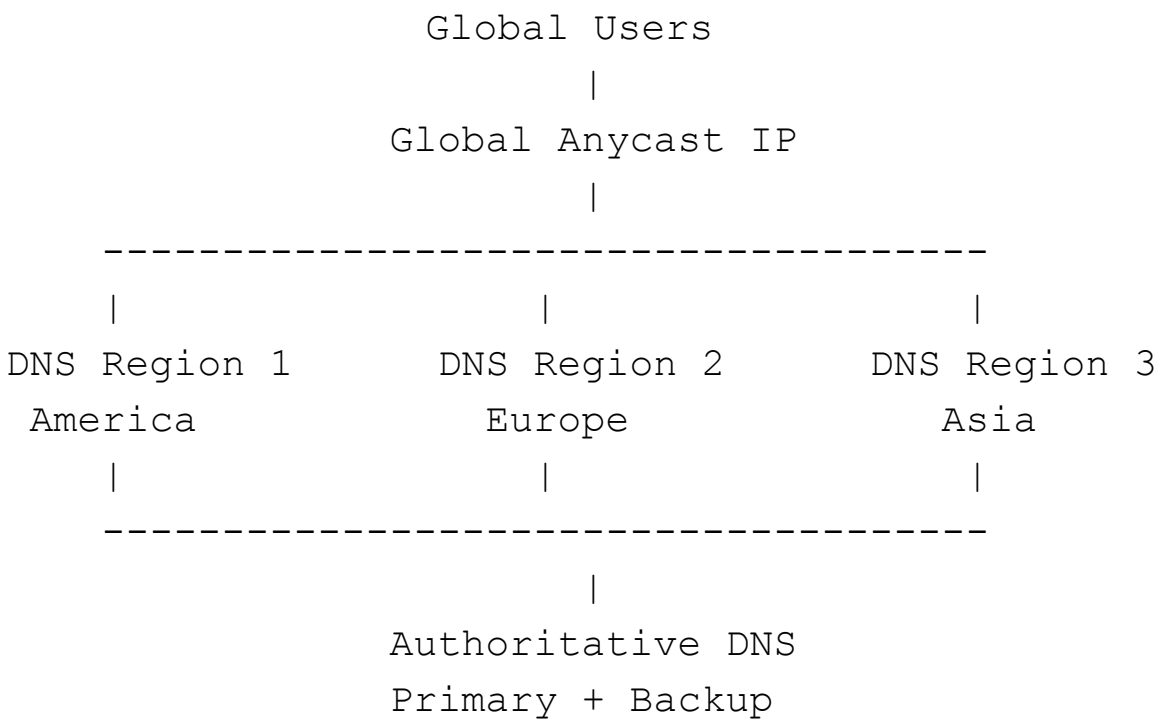
- High DNS lookup latency.
- Increased application startup time.
- Heavy load on the primary DNS server.
- A single point of failure.
- Poor performance for geographically distant users.
- Reduced scalability during promotional events.
- Increased probability of DNS congestion.

For millions of worldwide users, a single DNS server cannot efficiently handle sudden traffic spikes.

Therefore, centralized DNS directly affects application performance because users must wait for DNS resolution before connecting to the required cloud service.

5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.

A distributed architecture should use multiple DNS servers located in different geographic regions.



Design features:

1. Deploy DNS servers in multiple geographic regions.
2. Use **Anycast routing** to direct users to an appropriate DNS location.
3. Use DNS caching to reduce repeated queries.

4. Maintain primary and backup authoritative DNS servers.
5. Use suitable TTL values.
6. Monitor DNS server health continuously.
7. Automatically redirect traffic when a DNS location becomes unavailable.

This removes the single point of failure and reduces DNS latency for global users.

6. Analyze how TTL values, Anycast routing, and DNS caching influence performance and fault tolerance.

TTL:

TTL determines how long DNS records can remain cached.

- Higher TTL → fewer DNS queries and better performance.
- Lower TTL → faster record updates but more DNS queries.

Therefore, TTL should be selected according to the frequency of DNS changes.

Anycast routing:

Anycast allows multiple DNS servers to use the same IP address. Routing directs users toward an appropriate DNS location.

Benefits include:

- Lower latency.
- Better global performance.
- Traffic distribution.
- Improved fault tolerance.

DNS caching:

Caching stores previously resolved DNS records.

Benefits include:

- Faster subsequent lookups.
- Reduced DNS server workload.
- Reduced network traffic.
- Better scalability.

Conclusion: Combining **Anycast routing**, **appropriate TTL values**, and **effective DNS caching** provides both high performance and fault tolerance.

Q4.TCP Latency in Stock Trading

1. Analyze three possible TCP-related factors responsible for the latency increase.

The three major factors are **flow control, Nagle's algorithm, and SYN queue limitations.**

A. Flow Control

TCP uses a receiver window to prevent the sender from overwhelming the receiver.

If the receiving application cannot process data quickly enough, the advertised receive window may become small. This can restrict the sender and increase transmission delays.

In a high-frequency trading environment, even small delays can significantly affect transaction processing.

B. Nagle's Algorithm

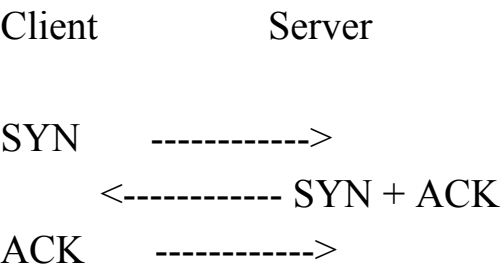
Nagle's algorithm combines small TCP packets into larger packets to improve network efficiency.

However, stock trading systems often transmit many small messages that require immediate delivery. Nagle's algorithm can introduce additional waiting time before a small packet is transmitted.

This can contribute to the increase from **1 ms to 40 ms** acknowledgment latency.

C. SYN Queue Limitations

TCP connection establishment requires the **three-way handshake**:



During peak market activity, many simultaneous connection requests can fill the server's SYN queue.

If the queue becomes overloaded, connection establishment can be delayed or requests may be dropped, increasing latency and retransmissions.

2. Explain how each factor affects transaction processing in high-frequency systems.

TCP Factor	Effect on Trading System
Flow control	Restricts sending when receiver buffers are limited
Nagle's algorithm	Can delay small transaction messages
SYN queue limitation	Delays or drops new connection requests
TCP retransmissions	Increase transaction acknowledgment time
Connection establishment	Adds additional delay for new connections

In high-frequency systems, transactions may arrive thousands of times per second. Therefore, even a small TCP delay can accumulate and affect order acknowledgments and customer experience.

The observed **increased TCP retransmissions, longer connection establishment time, and excessive buffering** indicate that TCP configuration and network congestion need to be addressed.

3. Recommend appropriate TCP configuration changes.

The following changes can reduce latency:

1. **Disable Nagle's algorithm** using `TCP_NODELAY` when immediate transmission of small trading messages is required.
2. Increase appropriate **TCP receive/send buffer sizes** to prevent unnecessary flow-control limitations.
3. Increase the **SYN backlog/queue capacity** so that more connection requests can be handled during market-opening bursts.
4. Use **persistent TCP connections** instead of repeatedly establishing new connections.
5. Monitor and reduce packet loss to minimize retransmissions.
6. Apply suitable network QoS for trading traffic.
7. Monitor TCP connection establishment and retransmission statistics.
8. Optimize server-side application processing so receive buffers do not become unnecessarily full.

Conclusion: For a high-frequency trading platform, minimizing unnecessary buffering, avoiding delays from Nagle's algorithm, increasing connection-handling capacity, maintaining persistent connections, and reducing packet loss can significantly improve transaction latency and reliability.